ASSET LOADING · SCENE SETUP

Choose a blue sky backdrop and delete the default cat

STEPS TO FOLLOW

1. Open Scratch → right-click cat → Delete
2. Choose a Backdrop → search "Blue Sky"
3. Click to add → stage turns sky blue!

💡 TIP

Any bright sky backdrop works. "Blue Sky 2" has clouds already painted in — saves you one step!

🤔 DID YOU KNOW?

Loading assets and configuring the scene before the game loop starts is called scene setup or initialisation — every game engine has this explicit phase.

OBJECT PROPERTIES · POSITIONING

Add a parrot or bird sprite, size it to 50, and position it on the left of the stage

SCRATCH BLOCKS — ON THE BIRD SPRITE

```
when 🏁 clicked
go to x: (-180) y: (0)
```

💡 TIP

Placing the bird at x=-180 gives it room to navigate. Pipes will come from the right side.

🤔 DID YOU KNOW?

Every game object starts with an initialization block that sets position, size, and velocity to known values before the game loop begins.

VARIABLES · PHYSICS SIMULATION

Create a "y velocity" variable; every frame subtract 1 (gravity) and apply it to y position

SCRIPT REFERENCE

SCRATCH BLOCKS — ON THE BIRD SPRITE

```
when 🏁 clicked
go to x: (-180) y: (0)
set [y velocity] to (0)
forever
  change [y velocity] by (-1)
  change y by (y velocity)
  if <(y position) < (-170)> then
    set y to (-170)
    set [y velocity] to (0)
```

TIP

Try "change [y velocity] by (-2)" for stronger gravity — the game gets much harder immediately!

DID YOU KNOW?

Velocity accumulation under constant acceleration is Newtonian mechanics — F=ma in code. Every physics engine from Box2D to Unreal uses this exact loop every frame.

EVENT HANDLING · INPUT POLLING

Press SPACE to set y velocity to +8 — fighting gravity with a flap impulse

📜 SCRIPT REFERENCE

SCRATCH BLOCKS — SEPARATE SCRIPT ON BIRD

– separate script on BIRD →

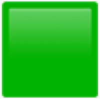
```
when [space] key pressed
set [y velocity] to (8)
```

💡 TIP

Try values like 6, 10, or 15 to find the feel you want. Higher = easier to stay aloft but harder to aim through gaps.

🤔 DID YOU KNOW?

"when [space] key pressed" is a hardware interrupt wrapper — the operating system tells Scratch the instant the key is pressed, with no polling delay.



OBJECT TEMPLATES · HIDING

Draw a green pipe sprite in the Paint Editor and hide it on flag click

📜 SCRIPT REFERENCE

SCRATCH BLOCKS — ON THE PIPE SPRITE

– PIPE: when 🏁 clicked →

```
hide
```

💡 TIP

Make the pipe width about 28–32 pixels. Too wide = impossible game. Too thin = too easy.

🤔 DID YOU KNOW?

The Pipe sprite is a class. Its clones are instances. OOP in action — one template, infinite objects.

REPEAT-UNTIL · DESTRUCTION

A pipe clone scrolls from right to left and deletes itself when offscreen

SCRATCH BLOCKS — ON THE PIPE SPRITE

— PIPE: when I start as a clone →

```
show
repeat until ‹(x position) < (-240)›
    change x by (-4)
delete this clone
```

💡 TIP

Increase the scroll speed to -6 or -8 for a harder game. This one number controls overall game difficulty!

🤔 DID YOU KNOW?

Parallax scrolling, endless runners, and side-scrollers all use this pattern: spawn off-screen right → scroll left → destroy off-screen left → repeat.

RANDOM NUMBERS · PROCEDURAL GENERATION

Make the master Pipe spawn clones at random heights in a forever loop

📜 SCRIPT REFERENCE

SCRATCH BLOCKS — ON THE PIPE SPRITE

– PIPE master: when 🏁 clicked →

```
hide
forever
    go to x: (240) y: (pick random (-60) to (60))
    create clone of [myself]
    wait (2.5) seconds
```

💡 TIP

Lower the wait to 1.5 secs for a rapid-fire challenge. Wider pick-random range creates more extreme height differences.

🤔 DID YOU KNOW?

Procedural generation — Minecraft biomes, Spelunky levels, No Man's Sky planets — is all "pick random" at different scales.

COLLISION DETECTION · GAME OVER

Detect when the Bird touches a Pipe — stop the game immediately

SCRATCH BLOCKS — INSIDE THE PIPE SCROLL LOOP

(inside the repeat-until scroll loop)

```
change x by (-4)
if <touching [Bird]?> then
    stop [all]
```

💡 TIP

Before "stop [all]", add "say [Game Over! 💥] for (1) secs" on the Bird sprite for a dramatic ending.

🤔 DID YOU KNOW?

"touching [Bird]?" checks pixel-by-pixel overlap 30 times per second — the same AABB collision test behind every hitbox, trigger zone, and collision mesh in game development.

VARIABLES · SCORE STATE

Add a Score variable that increments each time the bird passes a pipe

📜 **SCRIPT REFERENCE** Count the Score

SCRATCH BLOCKS — INSIDE THE PIPE CLONE SCROLL LOOP

(inside the pipe clone scroll loop)

```
if <(x position) < (-180)> then
  change [Score] by (1)
  wait (0.1) seconds
if <touching [Bird]?> then
  stop [all]
```

💡 TIP

Share your game on Scratch — click "Share" at the top and challenge friends to beat your high score! 🏅

🤔 DID YOU KNOW?

Threshold-crossing detection ("x < -180") is used in games, finance (stop-loss triggers), IoT (sensor alarms), and analytics (funnel drop-off).